MILITARY SPECIFICATION SHEET

FILTER ELEMENT, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 5 MICRON ABSOLUTE, P3 AIRCRAFT, RETURN, 14 GPM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-8815.

REQUIREMENTS:

Envelope: See Figure 1.

Pressure Drop: Maximum pressure drop at rated flow (14 GPM) across the clean filter element shall not exceed 3 psid at 100 ±10°F.

Contaminants: See Table I.

Degree of Filtration: The filter element shall remove 100 percent of all contaminants larger than 5 microns and 95 percent of all test contaminants. The filter efficiency shall be a minimum of 95 percent.

Element Collapse: The filter element shall withstand a differential pressure of 300 psid.

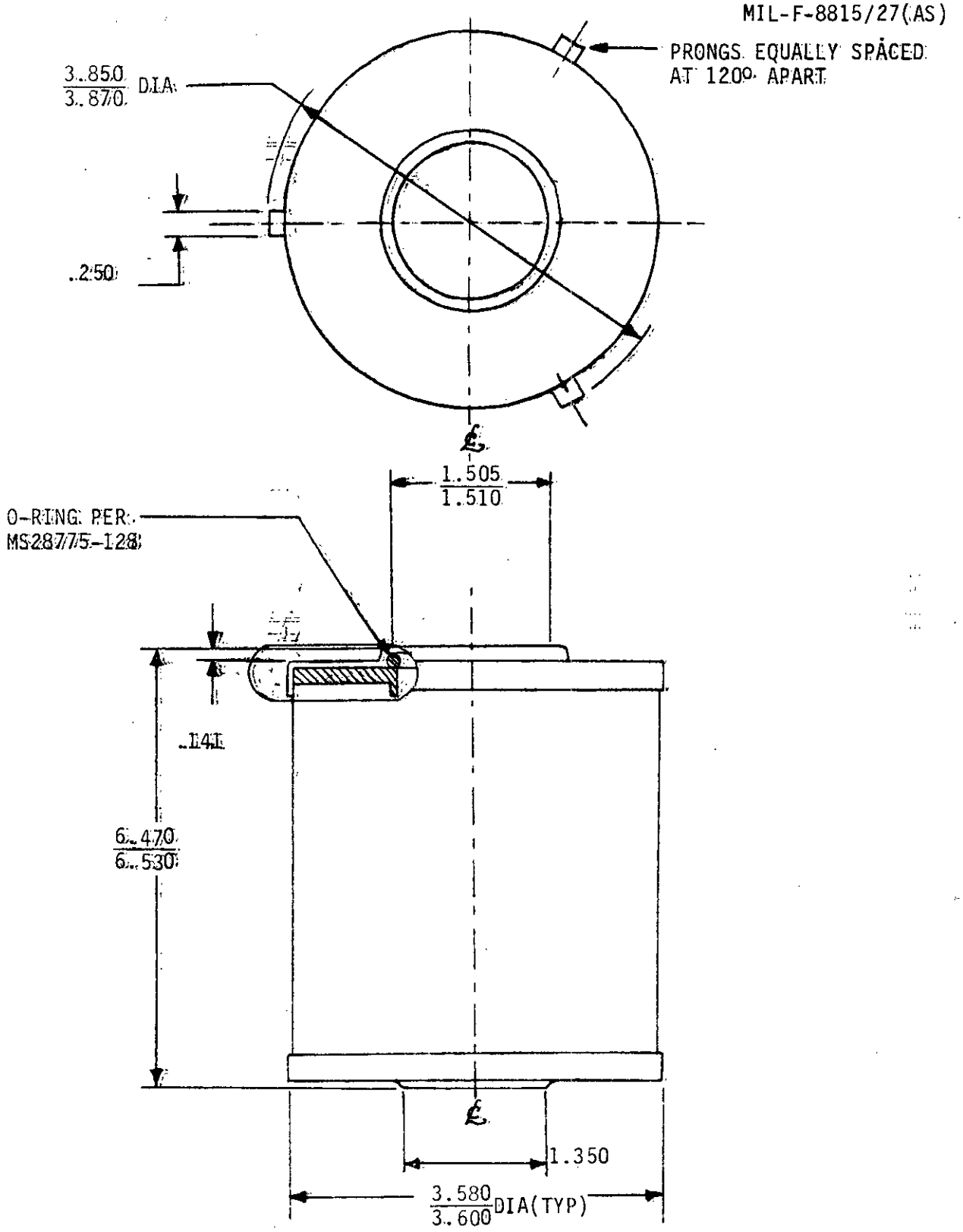
Cold Start: The element shall withstand 100 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop following the cold start test shall meet the requirement for pressure drop as shown above.

Maximum Particle Passed: The maximum particle passed shall be no greater than 5 microns.

FSC 1650

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NOTES:

1. Dimensions in inches unless otherwise specified.
2. Tolerances: decimals: ±.010; angles: ±1°.
3. Finish all over 100 except as noted. Surface texture in accordance with ASA B46.1 – 1962 (ANSI).
4. Material shall be in accordance with the procurement specification.

FIGURE I. DIMENSIONS.

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TABLE I. CONTAMINANTS

Test Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup (at 100 ± 10°F)		Migration Analysis Value Maximum Allowable (GMS)	
	F-9 Beads (GMS)	AC Dust (GMS)	F-9 Beads (GMS)	Carbonyl Iron E (GMS)	Add Rate AC Dust (GMS)	Minimum AC Dust to 20 PSID (GMS)	After Cold Start 4.7.2.3	After Vibration 4.7.2.8.3.2
14	0.2	0.2	0.05	0.05	2.0	20.0	-	.0085

Flow Fatigue: The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cpm maximum.

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
500 cycles at 3,000 psid pressure drop

Part Number: The part number shall consist of the basic number of this specification sheet.

Example:

M8815/27(AS)

Maximum weight: The maximum weight of the filter element shall be 1.5 pounds.

Intended Use: The filter element covered by this specification sheet is intended for use as specified in paragraph 6.1 of the general specification.

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TESTS:

The following tests shall be conducted on filter element number 1, in addition to the tests specified in the general specification. (See Table I of the general specification).

MIL-H-83282 Immersion: Following the immersion test specified in the general specification, the filter element shall be immersed in hydraulic fluid conforming to MIL-H-83282 for a period of 72 hours at a fluid temperature of 160°F and 72 hours at a fluid temperature of -65°F. All internal parts shall be in contact with the test fluid during this period. After immersion, the filter shall remain in this fluid at room temperature until further tests are conducted.

Element Bonding Test: Following the flow fatigue test specified in Table I of the general specification the bonding test shall be conducted to verify the element assembly integrity and whether degradation has occurred during prior testing. Support the element end cap (closed end) by means of a rigidly mounted fixture. The fixture shall distribute the load around the element end cap periphery. Apply an axial tensile load to the header (open end) for 15 seconds and then gradually reduce the load to zero. The load shall be 25 $^{+5}_{-0}$ pounds. Repeat the load cycle five times. After five cycles inspect the element. There shall be no element deformation or bonding failures.

Preparing Activity

Navy - AS

(Project No. 1650-N127)

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